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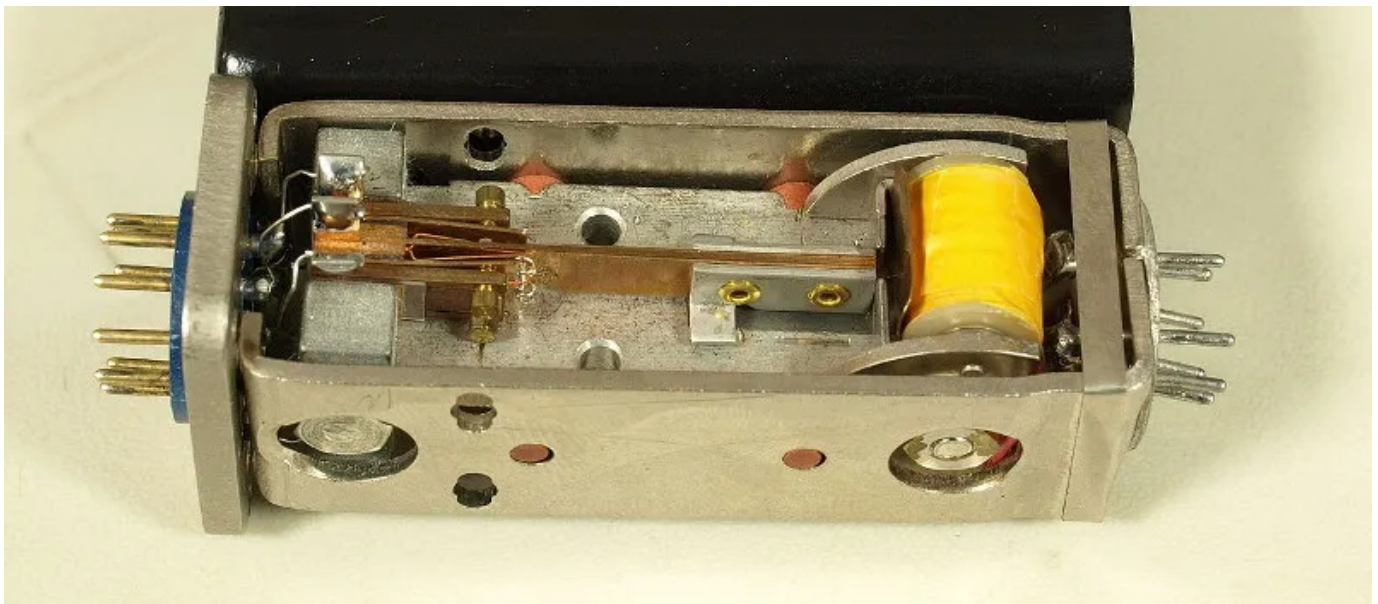
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CHOPPER AND CHOPPER-STABILISED AMPLIFIERS, WHAT ARE THEY ALL ABOUT THEN?

by: **Jenny List**

21 Comments

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One of my first jobs as a freshly minted graduate engineer involved the maintenance of a set of analogue chart recorders. They were museum pieces by the early 1990s: a motorized roll of graph paper across which a pen would traverse in proportion to the voltage on the input terminals. Inside was a simple servo, with a differential amplifier comparing the feedback via a potentiometer from the mechanism with the amplified input.

The recorders dated from the early 1960s and internally their electronics were from

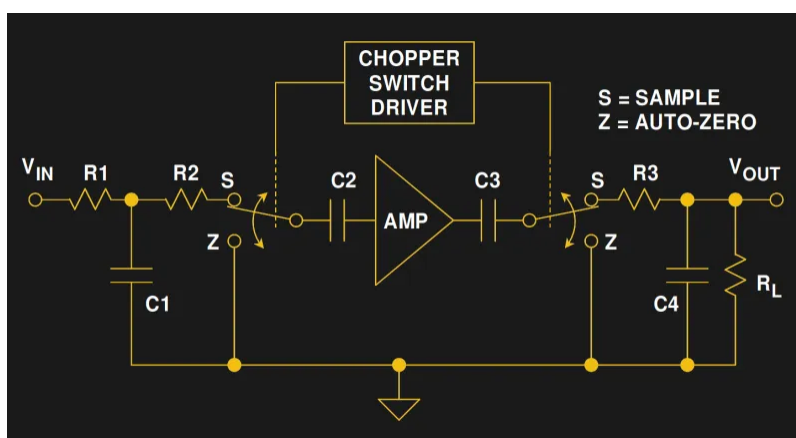
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Hz AC supply with a reed switch through its middle, something completely new to an overconfident youngster who thought she knew everything.

What I'd stumbled upon was a chopper amplifier, a slightly ungainly and long superseded solution to the problem of DC amplification from the days before ubiquitous integrated circuit op-amps. We have become so used to DC amplifiers that *just work*, that we have forgotten that there was a time when such devices were an impossibility. The close matching of properties between devices on the same wafer allowed integrated circuit op-amps to achieve stable DC amplification in a way that the best attempts at the same circuits with discrete transistors had failed, but before they happened some desperate measures were called for.

The chopper amplifier relied on the simple premise that, using 1950s technology, it was difficult to make a good DC amplifier but easy to make a good AC amplifier. Thus a DC input signal was modulated onto an AC source, fed through an AC amplifier, then demodulated and passed through a low-pass filter to recover the amplified DC signal. In my chart recorders the modulating and demodulating was performed by the 50 Hz reed switch, and the germanium transistor AC amplifier was able to do a good job of working with the resulting 50 Hz square wave.



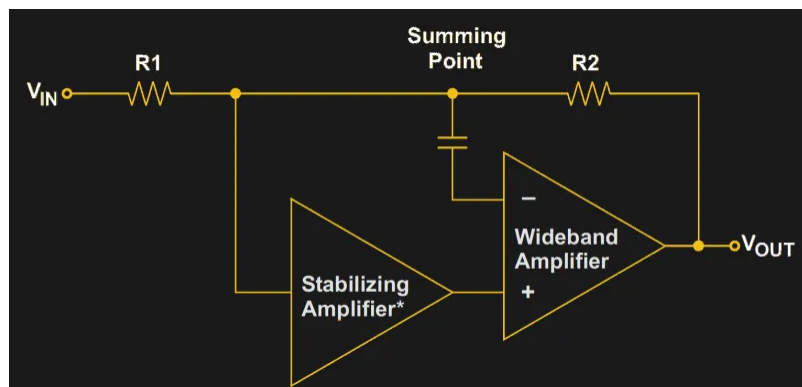
The classic chopper amplifier. From Analog Devices, MT-055 tutorial.

Chopper amplifiers were commonly found in instrumentation and other applications through the period until the advent of affordable integrated circuit op-amps through the 1960s. When these components arrived, they provided a single component that was a differential amplifier rather than the chopper's single-ended amplifier, with better gain and a hugely higher bandwidth at a fraction of the price and with none of the complexity. You might imagine that the chopper would be consigned to history then by this development, but that's not so! An integrated circuit op-amp is

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an op-amp with a low noise level at fractions of a Hz, i.e. at near-DC frequencies.

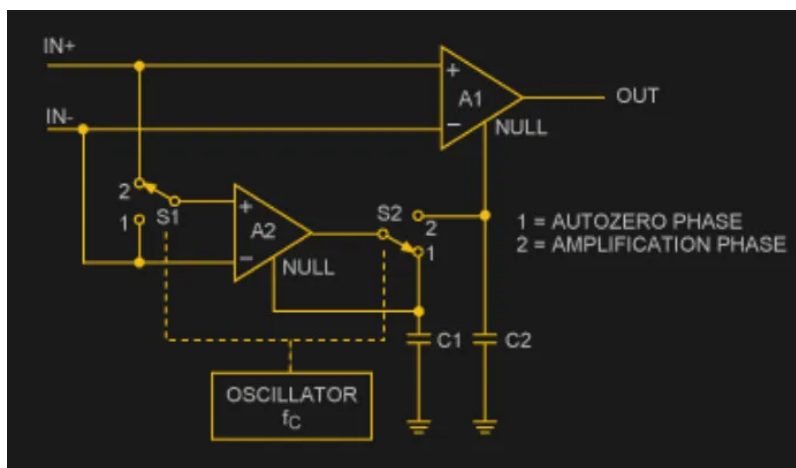


A chopper-stabilised amplifier, from the Texas Instruments app note: [Auto-zero amplifiers ease the design of high-precision circuits.](#)

The solution to the problem of poor op-amp DC noise performance came from a return to the chopper, but only as an adjunct to the op-amp rather than its replacement. The chopper-stabilised op-amp is a standard op-amp configured as an AC amplifier with a traditional chopper amplifier (labelled “Stabilizing amplifier” in our diagram) in the path to its non-

inverting input. This becomes a two-path device, with the high-frequency components being amplified by the op-amp with its superior bandwidth, but the DC component being amplified by the chopper amplifier for which the op-amp becomes simply a buffer. The whole circuit is still a single-ended one, but it has become a high-bandwidth amplifier with an ultra-low-noise DC performance.

There is a final refinement of the chopper amplifier, and it is one which can be bought off the shelf from multiple manufacturers. The auto-zero amplifier uses a chopper amplifier and an op-amp just as the chopper-stabilised amplifier does, except the chopper does not contribute directly to the output. Instead it forms a pair of sample-and-hold amplifiers, alternately trimming its own offset as a differential amplifier and then trimming the offset of the other op-amp which does the amplification. This has the effect of



A basic auto-zero amplifier, from [Maxim Integrated app note 4179.](#)

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It is unlikely that any of you reading this piece will deal with vintage chart recorders for a living, though the Hackaday readership never cease to surprise us. So if you encounter a chopper amplifier it's likely to be of the final variety, an auto-zero amplifier chip that will probably appear to you to be as just another op-amp that you're using for a strain gauge or similar. But even if the joys of maintaining a 1950s vibrating reed switch will never bring a wrinkle to your brow, it's still worth knowing something about how they work. For further information I'd suggest your next port of call should be the multiple application notes and data sheets on the subject that can be found on the web or the few linked from this page. You can never have too many pieces of circuitry stored in your brain!

Image: a mechanical vibrating switch used in a vintage chopper amplifier.

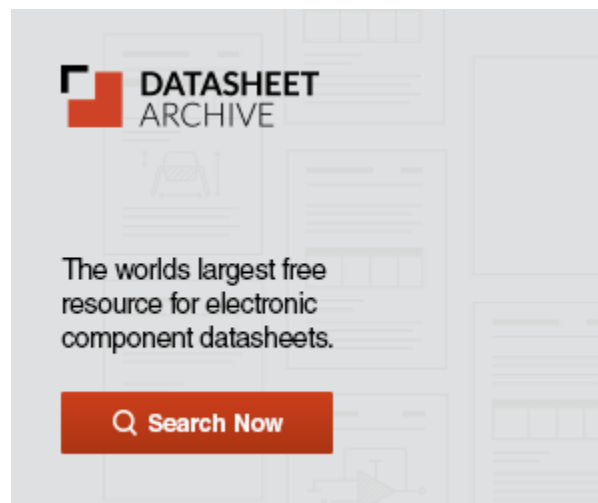
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